

Conceptual and practical issues limit the utility of statistical estimators of phenological events

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Abstract. Widespread shifts in phenological events in response to climate change have inspired phenological monitoring programs and new methods for analyzing sparse phenological data. For example, the Weibull distribution is increasingly used to estimate the dates of hard-to-observe phenological events, such as first and last flowering dates, in sparsely or unsystematically sampled data sets. In contrast, recent application of the Weibull estimator to an intensely and systematically sampled flowering phenology data set unexpectedly found different results than a previous analysis of the observed dates; at issue is whether different aspects of phenological curves shift uniformly or disparately. We used this case study to (1) raise conceptual and technical issues around when and how to infer phenological events using Weibull (or other) estimates, and (2) re-analyze the data set in question with these considerations in mind. Our re-analysis using the Weibull estimator shows that first, peak, and last flowering dates shift disparately through time, supporting the original analysis of the observed dates. We show that off-the-shelf usage of statistical estimators to generalize about an unsampled population may be inappropriate without considering how well sampled the focal study population is and how biological features such as habitat heterogeneity influence the natural scope of the unsampled population.

Key words: climate change; flowering phenology; habitat heterogeneity; phenology; sampling; Weibull distribution.

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INTRODUCTION

Shifts in the timing of life cycle events—phenological events—in response to changing climatic conditions have become bioindicators of climate change (de Groot et al. 1995, Menzel 2002, Orlandi et al. 2005). The importance of phenology as a biological response to climate change has inspired phenological monitoring programs and new methods for analyzing phenological data, such as those

from historical records collected by naturalists, herbarium and museum samples, or community (citizen) science (e.g., the USA National Phenology Network, the European Phenology Network, Climate Watch by the Audubon Society, and Budburst). While these data sets can help reveal historical and present-day patterns of phenological variation, doing so requires that the often sparse and unsystematic records in such collections be analyzed with these limitations in mind.

The Weibull distribution is increasingly used to estimate the dates of hard-to-observe phenological events, such as first and last flowering dates (Pearse et al. 2017, Belitz et al. 2020). Applying

[Correction added on 15 January 2022, after first online publication: The US state of Maryland in affiliation 4 has been corrected to Massachusetts. This updated version of the article corrects this mistake.]

such an estimator to historic, museum-based, and community science data sets is logical because records within such data sources are unsystematically collected and sparse in their time resolution. However, it is less clear what one gains from applying a statistical estimator to a permanent, intensively sampled phenology study, where there is little uncertainty in first or last flowering dates; in such data sets, dates derived from statistical estimators should closely resemble the observed dates. Therefore, any discordances in the results from analysis of dates with and without applying a statistical estimator should be interpreted with caution. As statistical estimators for phenology gain popularity, it is increasingly important to understand the consequences of their use across data contexts.

A recent analysis that applies the Weibull statistical estimator to a sampling-intensive long-term data set from Colorado, USA, finds that first, peak, and last flowering dates shift uniformly across plant species (Pearse et al. 2017). In contrast, a previous analysis of the same data set based on observed (unestimated) dates finds that these three aspects of phenology shift disparately (CaraDonna et al. 2014). Pearse et al. (2017) claim that these contrasting results occur because of sampling biases in the raw data, which the statistical estimator corrects by providing estimates for the true statistical population. However, these sampling biases are neither defined nor described, and it is unclear why analysis of estimated dates would produce different results than analysis of observed dates for such an intensely sampled data set. Here, we use this case study to (1) raise conceptual and technical issues around when and how to infer phenological events using Weibull (or other) estimates, (2) re-analyze the Colorado data set with these considerations in mind, and (3) conclude by discussing when the use of statistical estimators is warranted.

Conceptual and technical issues when preparing data for a statistical estimator

The data set in question is from the Rocky Mountain Biological Laboratory, Colorado, USA. The number of flowers is counted for each species in bloom in a series of permanent 2×2 m plots approximately every other day throughout the flowering season, from 1973 to the present (Inouye 2008, CaraDonna et al. 2014). Thus, we

know within ± 1 d when each species began to flower and stopped flowering within each plot, for the vast majority of cases. Exceptions include a small number of cases in which researchers arrived after some early-season species already began to bloom (see CaraDonna et al. 2014 for discussion of this point). Plot-level flower counts may come from a single plant or multiple plants. The farthest distance between any two plots is ~ 1 km, and the largest elevation difference is 106 m (Inouye 2008); most species are sampled in plots that are separated by as little as ~ 5 m and up to 500 m from one another.

Prudent application of the Weibull estimator to a data set such as this might look something like the following. Flowering plots are independent and representative samples from the larger population whose natural bounds are imprecisely known, but within each plot, the flowering dates for each species are directly, and precisely, observed. We would like to use these data to generalize about the possible range of dates on which an earlier and unobserved date of first flowering might occur in the larger, unsampled population. To produce such an estimate, we would use the distribution of dates with nonzero flower counts, starting with the first observed flowers in each plot, to produce a single estimate for a given year for each focal plant species. Assuming that the distribution of possible earlier dates for the unsampled population is approximated by a Weibull distribution, we can use the data from our plots to model the behavior of the early tail of the flowering time distribution by backward extrapolating to a maximum-likelihood value of earliest flowering date given the data. Our estimate should be always at least as early as the earliest observed date and should occur later than strictly bounding biological phenomena, such as the date of snowmelt in this case.

In the remainder of this section, we discuss how the approach of Pearse et al. (2017) differs from the scenario above, by (1) generating multiple estimates for the unsampled population for each species in each year, (2) replicating a date for every single flower count in the data set when estimating flowering dates, and (3) omitting biologically informed limits to phenological events to avoid unrealistic date estimates. Each of these three factors obscures the value of Weibull estimates for phenological data.

Multiple estimates for the unsampled population.—By applying the Weibull estimator to each plot, Pearse et al. (2017) generate multiple estimates of the true statistical population for each species in each year. This choice contradicts the premise of generating a population-level estimate from replicate sampling units, where each plot serves as a *sample* that informs the population-level estimate within each year. When the estimator is applied to dates within each 2 m × 2 m plot, estimates are created for dates that are known with near certainty (off by no more than 1 d in the vast majority of cases). Performing plot-level estimates creates several problems for downstream analysis and interpretation. First, the estimated dates for some plots can be *later* than the first observed flowering date for other plots within the larger sample (and therefore later than the observed onset date in the sample population). Second, rather than choosing the earliest of the estimated dates across plots for a given year for each focal species, Pearse et al. (2017) fit a line through all the plot-level estimates within each year, which means the rate of change through time (the slope of the line) is based on the *average* of these plot-level estimates. Because the average onset date will always be later than the earliest estimate, Weibull estimates are systematically biased upward: 46% of the average estimated dates occur *later* than what is actually observed in the sample population (Fig. 1A). Thus, it is important to carefully consider the sampling scheme when applying estimators to phenological data, to ensure that estimated first flowering dates for the unsampled population are earlier than the observed dates (or later in the case of the flowering end).

To further illustrate these points with an example, consider flowering onset for the first species alphabetically in the species list in the first year of the study, *Achillea millefolium*, which flowers near the end of the season, in 1974. *Achillea millefolium* is present in five plots. After applying the Weibull estimator to each plot, the average estimated onset date across all five plots is 19 July. This estimate is problematic because the earliest flower in the sample population was *observed* on 12 July. The estimated onset date calculated correctly—at the population level—for this species in 1974 is 9 July, slightly earlier than the observed flowering onset date, as expected (Fig. 1B).

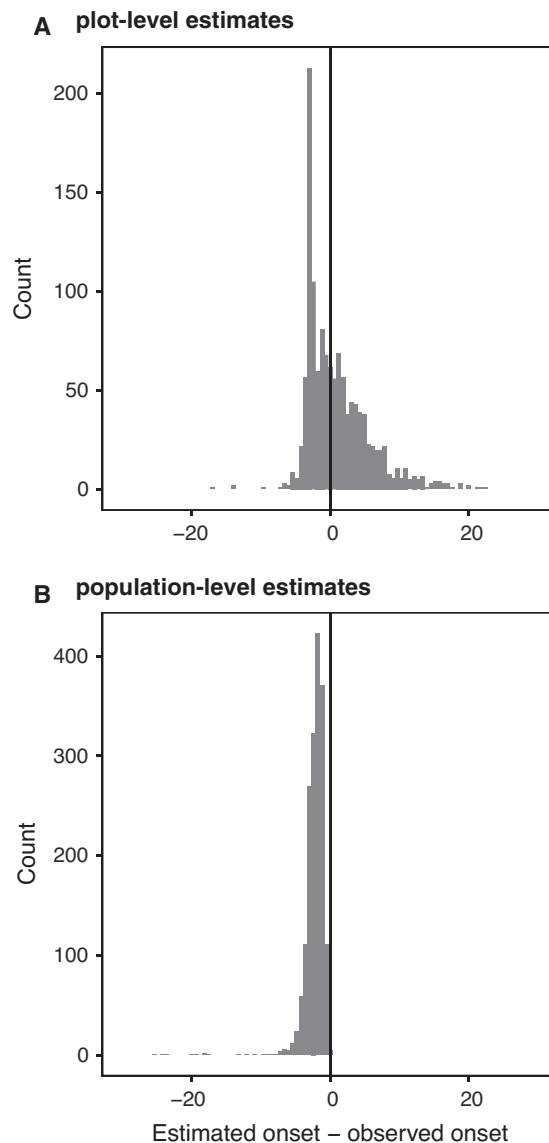


Fig. 1. Histograms showing the effect of estimating onset dates at (A) the plot level vs. (B) the population level, using unreplicated input dates (results are the same using replicated dates; Appendix S1: Fig. S1). The x-axis is the estimated flowering onset date minus the observed flowering onset date. Values to the left of 0 indicate estimated dates that are earlier than observed dates, as expected when using a statistical estimator to estimate dates for a true population.

Date replication.—Pearse et al. (2017) additionally assign a date to every single flower count in the Colorado data set, although this replication of dates is never mentioned in the text, nor is it

included in an associated correction (Pearse et al. 2019). For example, if a species has 25 flowers on 15 May in a given plot, 15 May is replicated 25 times in the input to the Weibull calculation (for the given plot). If the same species has 35 flowers in the same plot at the next census on 17 May, 17 May is replicated 35 times. Applying this same logic of abundance-based date replication to a herbarium record, a specimen containing ten flowers would be assigned ten dates. These replicated dates then go on to form the data that the Weibull estimator uses to estimate onset or end. This date replication represents a methodological choice and not a limitation of the Weibull estimator per se. But because one cannot distinguish between true individuals vs. flowers in the Colorado data set, replicating dates by flower counts will induce systematic differences between species with different average display sizes. The most important consequence of date replication in the Colorado data set is that replicated plot-level estimated onset dates are on average later than non-replicated dates (Appendix S1: Fig. S2A), resulting in another systematic, upward bias in the results presented in Pearse et al. (2017, 2019).

In contrast to the plot level, replicating dates has no systematic bias on date estimates calculated at the population level (Appendix S1: Fig. S2B). In fact, when removing the plot-level stratum and instead performing the estimation at the population level, replicating dates in the manner described above does not affect the original conclusion of the analysis of the observed dates, namely, that distinct aspects of phenological curves shift disparately through time (Appendix S1: Table S1; CaraDonna et al. 2014). Importantly, this conclusion differs from that made by Pearse et al. (2017, 2019) using the same data and the same underlying Weibull calculations supplied in the associated R package *phest* (<https://github.com/willpearse/phest/>). We elaborate on this point further below when discussing our re-analysis of the Colorado data set (*Methods*).

When applying the calculations implemented in *phest* to the Colorado data set *without* date replication, one loses the information about daily flower abundance and instead collapses each record to a binary presence/absence species indicator at the plot level. While this sacrifices abundance data, it nonetheless allows a statistically

sound estimation of dates from the Colorado data set. Furthermore, data on flower abundance are not necessary for defining the onset of an event when plots are sampled frequently. First flowering date is the date on which flowers are first observed, regardless of floral abundance on that date (vice versa for last flowering). By definition, the onset of flowering occurs when flowers first open and when they have not been open on a previous census, which is known within one day of error in the vast majority of cases in the Colorado data set.

Additionally, our re-analysis (*Methods*) incorporates flower count data in two important ways. Firstly, peak flowering dates, which are not estimated because they are not distributional extremes, are defined based on flower abundance as the date by which 50% of total season flowers for a given species were counted (*following* Iler et al. 2013, CaraDonna et al. 2014). Secondly, following Pearse et al. (2017), flower abundance data are directly incorporated into downstream analyses that estimate rates of change through time in flowering dates, to control for any effects of population size on phenology (Miller-Rushing et al. 2008).

Biological bounds on date estimates.—Next, incorporating biologically informed limits to phenological events will avoid unrealistic date estimates. For example, the earliest estimated onset date in the data set of Pearse et al. (2017) is the day of year −690 (flowering into previous years). This is clearly incorrect. Fortunately, such unrealistic dates compose a tiny fraction (<1%) of the Weibull-estimated onset dates in Pearse et al. (2017). Nonetheless, such dates can more generally interfere with parameter optimization and affect overall model fit. Therefore, we encourage researchers to set limits to date estimates that are based on biological reality, or to have an explicit outlier detection routine.

One additional factor to consider is biological limits to dates for life history events in the study system of interest. For example, in the study area of the Colorado data set, flowering does not commence until after the initial spring snowmelt. Therefore, any estimated date prior to snowmelt is incredibly unlikely. Other study systems may have known photoperiod requirements before flowering can commence. At the least, for temperate ecosystems in the Northern Hemisphere

with a defined winter season, researchers could set a limit to prevent estimates from going into previous years. Such limits to estimated dates should be simple to incorporate into the Weibull estimation procedure and would make its application stronger. Indeed, we check for implausible estimates in our re-analysis (*Methods*). Additionally, applying sensible, biologically informed hard bounds on date estimates may help reveal where extreme-value distributions such as the Weibull may be inappropriate for the estimation problem at hand.

METHODS

We apply the Weibull estimator following the criteria in Pearse et al. (2017): At least three unique dates and more than 10 dates are required to be included in the Weibull estimate calculation, and no more than 30 dates are used (i.e., $\max k = 30$). Because flowers are counted every other calendar day throughout the flowering period, requiring more than 10 dates means that species that flower for <22 d in a given year are excluded from the Weibull estimate calculation. This leaves 52 species compared with 60 species in CaraDonna et al. (2014); this is why the species list differs between our analysis here and that in CaraDonna et al. (2014). We additionally exclude species with fewer than 19 yr of data and the entirety of the initial year of data collection (1973), which included only a subset of the plots (following Aldridge et al. 2011, CaraDonna et al. 2014).

We apply the Weibull estimator to flowering dates at the population level for each species in each year to estimate first and last flowering dates. With this approach, flowering date values populate the sample for the estimator as different plots come into flower and as existing plots continue to flower. Because of variable flower abundance across plots and the well-known sampling effect of flower abundance (i.e., population size) on phenology (Miller-Rushing et al. 2008), CaraDonna et al. (2014) sum flowers across plots to create a population-level flowering curve for each species in each year. Their goal is to derive population-level flowering dates that are comparable to the majority of long-term phenology data sets, which are limited to single values in each year for each species (typically the first date

of appearance). Thus, our approach for applying the Weibull estimator provides a consistent comparison with the observed dates in CaraDonna et al. (2014), because our approach results in a single date per species per year inputted into the analysis of shifts in flowering dates through time.

We next analyze rates of change through time in these estimated flowering dates and in peak flowering date. Peak flowering date is not estimated because it is not a distributional extreme. Following CaraDonna et al. (2014), peak flowering date is defined as the date on which 50% of flowers are counted. Following Pearse et al. (2017), we fitted linear models with estimated flowering date (or peak flowering date) as a continuous response variable, species as a categorical predictor, year as a continuous predictor, a species $\times$ year interaction term, and log-transformed peak flower abundance as a continuous predictor. The flower abundance predictor in these statistical models accounts for potential effects of population size on phenology. To preserve intervals between each year and to obtain slope estimates with per-year units, we subtracted the baseline year value such that the time series is inputted as 0, 1, 2, ..., t years, where $t \geq 19$.

We then asked how shifts in first flowering predict shifts in peak and last flowering. To do this, we extracted the maximum-likelihood estimate (and standard error) of the rate of change for each species across the time series of each phenological event (onset, peak, end) from the linear models described above. Estimates, and their errors, for each pair of phenological contrast (onset vs. peak, onset vs. end) were inputted into Deming regressions (following Pearse et al. 2017). As described in CaraDonna et al. (2014), the vast majority of long-term flowering phenology data sets are limited to single dates of first appearance. CaraDonna et al. (2014) were therefore interested in how representative first flowering dates were for detecting shifts in other phenological events that are less commonly measured: timing of peak abundance and end of flowering. Following Pearse et al. (2017), we agree that Deming regression is appropriate in this context because it accounts for error in both predictor and response variables (Therneau 2018).

RESULTS

On average, across years for each species, the Weibull estimates that flowering onset occurs 2.29 ± 0.071 d earlier compared with the observed data, and 3.40 ± 0.26 d later for flowering end (mean ± 1 SEM, $n = 52$ species). Our Weibull-estimated first and last flowering dates are realistic, in that they occur within the growing season for a given year. For example, the earliest Weibull-estimated date in our entire data set is for the first species to flower, *Claytonia lanceolata*, and occurs on 30 March (day of year 88.8) in a single plot; the observed first flowering date in that plot and year (2004) is 5 April (the day of year 95). The snowmelt date for that plot was “earlier than April 4” (the plots were not visited prior to April 4), so we conclude that this estimate falls within a biologically reasonable time frame. Similarly, there were no flowering end dates that extended into the next year or that extended so late into the fall as to be biologically unrealistic.

Each of the linear models used to estimate shifts in first, peak, and last flowering dates has an adjusted $R^2 \geq 0.72$ (flowering onset date: $R_{adj}^2 = 0.76$, $P < 0.0001$; peak flowering date: $R_{adj}^2 = 0.81$, $P < 0.0001$; and flowering end date: $R_{adj}^2 = 0.72$, $P < 0.0001$; Appendix S1: Table S2). The Deming regressions using the phenological shifts from these linear models show that first, peak, and last flowering dates shift disparately. Specifically, change in first flowering date overpredicts change in peak flowering date and does not predict change in last flowering date (Fig. 2). For every day that first flowering advances, peak flowering advances by 0.77 d (95% CI: 0.58–0.96, does not overlap with 1; Fig. 2A). Change in first flowering does not predict change in last flowering (95% CI: -1.12 to 3.07 d, overlaps with 0; Fig. 2B). We also find that peak flowering underpredicts change in first flowering; for every day that peak flowering advances, first flowering advances by 1.30 d (95% CI: 0.98 to 1.61 d; Appendix S1: Fig. S3A). Although the 95% confidence interval for peak vs. first slightly overlaps with 1, the slope estimate is consistent with the result of change in first overpredicting change in peak, in that change in peak underpredicts change in first. Finally, change in peak flowering does not predict change in last flowering date

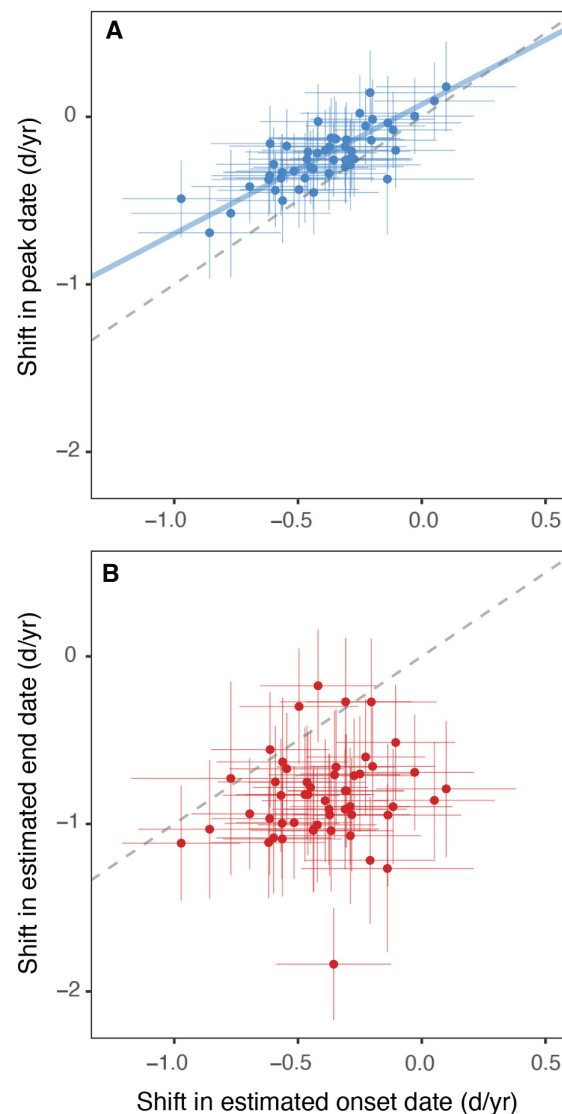


Fig. 2. Degree to which temporal shifts in Weibull-estimated first flowering dates predict (A) observed shifts in peak and (B) shifts in Weibull-estimated last flowering dates, from a phenological data set in the Colorado Rocky Mountains, USA ($n = 52$ species). Peak dates are not estimated because they are not distributional extremes. Each dot represents the rate of change in flowering date per year for a species, with associated standard error from linear models described in the main text. Dashed gray lines are 1:1 lines. Line of fit in (A) is from a Deming regression. In (A), the 95% CI does not overlap with 1, showing strong support that the slope is <1 , and in (B), the 95% CI overlaps with 0, making the slope indistinguishable from a flat line (see the main text for 95% CIs).

(95% CI: -1.69 to 6.52 d, overlaps with 0; Appendix S1: Fig. S3B).

DISCUSSION

Our re-analysis using the Weibull estimator supports the original conclusions of the analysis of the observed dates: Separate aspects of a population-level phenological response can indeed shift disparately through time, especially with regard to the end of an event (CaraDonna et al. 2014; sensu Carter et al. 2018). Using the Weibull statistical estimator, shifts in first flowering date overpredict shifts in peak flowering date and fail to predict change in last flowering date. Indeed, one would expect that application of a statistical estimator to an intensively sampled data set would come to the same conclusions as an analysis of the observed data. Disparate rates of change among different components of a population-level phenological curve are likely to arise if individuals within the population vary in the extent to which they shift the timing of their flowering in response to climate change (CaraDonna et al. 2014).

We now turn to a conceptual discussion of when and how to use estimators such as the Weibull more broadly. Specifically, we consider the implications of different spatial arrangements of sampling units, for what type of data an estimator is most useful, and alternative approaches for learning more about an unsampled population. For the Colorado data set, as already discussed, we argue that it is most conceptually and statistically sound to allow each plot to serve as a sample that informs the species-level estimate within each year. However, the approach for other studies will depend on the degree of spatial separation of the sampling units and the research question of interest. For example, one could argue for application of the Weibull estimator to each replicate sample if a series of plots were sufficiently far apart from one another as to be clearly sampling from different populations. In this scenario, one would be estimating a date for a relatively broad area based on a single sampling unit, so that $n = 1$ for each population. If such sparse information was all that was available, it would make sense to apply the statistical estimator to each plot, because that is the only sample of the biological population.

Additionally, an alternative approach for the 30% of species in the Colorado data set that occur in both a “lower” and an “upper” set of plots (sensu Aldridge et al. 2011) would be to calculate species-level estimates based on *each set of plots* separately. It is theoretically possible that the upper and lower plots could sample from different plant populations for this subset of species that occur in both. Thus, one would still allow each plot to inform a species-level estimate as we have done in our re-analysis, but there would be two separate estimates: one for the lower set of plots and one for the upper set of plots. We did not break up the plots into this level of detail here because our goal is to repeat an analysis for comparison between two previous studies. These considerations are applicable to any sampling scheme with replicate sample units. Researchers applying estimators such as the Weibull must therefore carefully justify the scope of the hypothesized true population(s) whose parameters are being modeled.

The Weibull estimator shows the most promise for data that are collected unsystematically instead of systematically, such as herbarium records, community science and crowd-sourced data, and some historical data collected by naturalists (Taylor 2019, Belitz et al. 2020). Indeed, historically the Weibull distribution has been applied to sparsely and/or unsystematically collected data (e.g., to estimate extinction dates; Pearse et al. 2017). In contrast, frequently and systematically collected phenology data from well-defined samples, such as the Colorado data used here, are most useful for validating an estimator. Application of the Weibull estimator to a subsample of phenological data from a population in which all individuals are censused would allow one to assess how close the estimated dates are to the true testing population. For example, Taylor (2019) use a data set in which all individuals of a species within an isolated habitat fragment are frequently censused to compare the accuracy of different estimators, and the Weibull performs well at the population level. One should also be able to predict the actual date of first observation if this observation is left out of the set of later dates used to generate the Weibull estimate. Additionally, the Weibull is also useful for filling in missing records in systematically collected data, as done here for estimating

flowering onset for a few cases where researchers missed the beginning of flowering. As expected, our re-analysis here also supports the conclusion that the Weibull can serve as a robust statistical estimator for population-level onset and end dates.

If one wishes to know more about the broader population than the sample population of an intensely sampled phenology project, one should consider landscape heterogeneity and its effect on phenology. If we think practically about the unsampled population in our subalpine study area, as the sampling area expands, landscape heterogeneity increases quickly, as does heterogeneity in abiotic conditions (temperature and snowpack) that are known to affect flowering phenology at this site (Inouye et al. 2002, Forrest et al. 2010, Anderson et al. 2012, Iler et al. 2013). To better understand phenology across a heterogeneous landscape, and therefore to learn more about the broader population, one should quantify climate–phenology relationships and how those relationships scale across the landscape. Although the Weibull-estimated dates are theoretically representative of the true statistical population, they do not explicitly take environmental heterogeneity into consideration nor help us to understand the natural bounds on the hypothetical “true” population.

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